

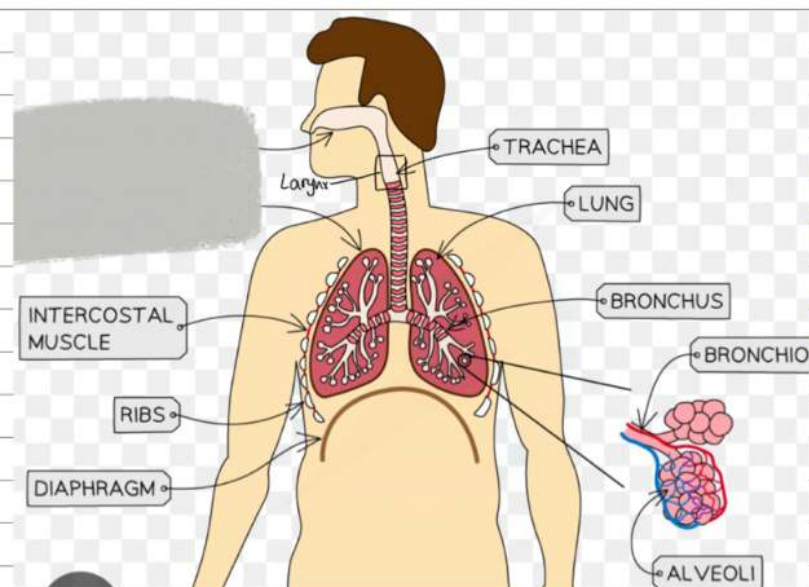
Gas exchange in
humans

Adaptations of the alveoli for gas exchange:

- Folded to increase surface area
- Have walls one cell thick to reduce the distance for diffusion
- Surrounded by a rich supply of blood capillaries for faster and more efficient diffusion.
- Have a moist surface as cells die if not kept moist.

Features of the respiratory surfaces:

- 1- Large surface area exposed to the gas: The presence of millions of alveoli in the lungs provide a large surface area for gas exchange.
- 2- Short distance diffusion: Two-layer thin epithelium; one layer is the alveolus wall, and the other layer is the capillary wall; for faster diffusion
- 3- Good blood supply: The alveoli is surrounded by networks of blood capillaries. The blood capillaries remove all the oxygen around, keeping low oxygen supply so that the diffusion gradient is maintained to keep on absorbing oxygen in the alveoli for cells. Carbon dioxide in the alveoli is removed by ventilation, releasing it outside of the body, this encourages the diffusion of carbon dioxide into the bronchioles.
- 4 Ventilation: The movement of air into and out of the lungs (Breathing action)



* Function of the ribs:

- Protect lungs and heart
- Move to ventilate the lungs

* Function of cartilage in the trachea:

- Prevents the trachea from collapsing
- Allows the trachea to expand and contract slightly during breathing and swallowing
- Protection

Comparing the carbon dioxide content of inspired and expired air:

1- Limewater is a test for CO_2 , when a high concentration of CO_2 is bubbled through Limewater, it turns from clear colorless to cloudy white.

2- Sodium hydrogen carbonate is a test for CO_2 . It changes color if the pH changes. It changes from red to yellow when CO_2 is bubbled through it as CO_2 dissolves in water to form carbonic acid which increases the pH.

	Inhaled/%	Exhaled/%
oxygen	21.00	16.00
carbon dioxide	0.04	4.00
water vapour	variable	saturated

- Used up in respiration
- Produced in respiration
- Produced in respiration
- Moisture evaporates from the surface of the alveoli

Why do we breathe faster and deeper during exercise?

Physical activity increases breathing rate as more oxygen is needed because muscle cells respire more to produce more energy which is needed in muscle contractions. CO_2 is produced as a by product of respiration and it dissolves in the blood plasma forming carbonic acid which lowers the pH of the blood. More CO_2 is produced so higher breathing rates are required to remove the CO_2 to maintain the pH and CO_2 levels in the blood.

Why do we keep on breathing faster and deeper after exercise?

During exercise, oxygen is not delivered to the muscles fast enough so muscles use anaerobic respiration to produce the energy needed for muscle contraction. Lactic acid is produced as a by product of anaerobic respiration, and it accumulates in the blood and muscles. It becomes toxic if not broken down and it causes muscle aches and cramps. Thus, breathing rates remain high even after exercise as extra oxygen is needed to break down the accumulated lactic acid into harmless substance in the liver. This process is called oxygen debt.

→ lactic acid + oxygen → Carbon dioxide + Water

Inspiration:

- 1- External intercostal muscles contract, pulling the ribcage upwards and outwards.
- 2- Internal intercostal muscles relax
- 3- Diaphragm muscles contract and the diaphragm moves downwards
- 4- Volume of the thorax increases and pressure in lungs decreases.
- 5- Air rushes into the lungs to equalize the pressure as the atmospheric pressure is higher

Expiration

- 1- External intercostal muscles relax pulling the ribcage downwards and inwards
- 2- Internal intercostal muscles contract to cause forced expiration
- 3- Diaphragm muscles relax and the diaphragm moves upwards
- 4- Volume of the thorax decreases and pressure in lungs increases
- 5- Air rushes out of the lungs to equalize the pressure as the atmospheric pressure is lower

Ciliated epithelial cells in the nose and trachea: Specialized cells that have hair-like structures on their surface called cilia

- Cilia sweeps away the mucus towards the stomach and away from the lungs
- Mucus is a thick watery fluid that traps bacteria and dust from the air
- Goblet cells: Produce mucus